

AN ℓ_1 -SUM OBSTRUCTION TO A DIRECTIONAL SMOOTHNESS PROPERTY

RUN FA_BANACH_001

STATUS

This is a candidate partial result for the question in Remark 2.11 of Dilworth–Garrigós–Hernández–Kutzarova–Temlyakov [1]. It does not solve the full question. It rules out a natural class of non-uniformly smooth spaces as possible examples.

SOURCE QUESTION

In Definition 1.12 of [1], a pair $(X, \mathcal{D})$ has property $\mathcal{D}(s, c_1)$ if, for some $s > 1$ and $c_1 > 0$,

$$\text{dist}(f, [\varphi]) \leq \|f\| (1 - c_1 |F_f(\varphi)|^s)$$

for every $f \neq 0$ and every atom $\varphi \in \mathcal{D}$, where F_f is an associated norming functional. Remark 2.11 asks whether this property could hold for some non-uniformly smooth Banach space and some dictionary.

8 S. DILWORTH, G. GARRIGÓS, E. HERNÁNDEZ, D. KUTZAROVA, AND V. TEMLYAKOV

COROLLARY 2.9. *Suppose that $\rho(t) \leq \gamma t^q$, for some $\gamma > 0$ and $q > 1$. Then, for every dictionary $\mathcal{D}$, the pair $(\mathbb{X}, \mathcal{D})$ has the property $\mathcal{D}(s, c_1)$ with $s = q'$ and some $c_1 = c_1(\gamma, q) > 0$.*

REMARK 2.10. Observe that $\rho(t) \leq \gamma t^q$ can only hold for $q \leq 2$, and hence $q' \geq 2$. However, property $\mathcal{D}(s, c_1)$ may hold in some cases with s close to 1. For instance, if $\mathbb{X} = \ell^p$ with $p > 2$ and $\mathcal{D}$ is the canonical basis, then it is not hard to verify that $(\mathbb{X}, \mathcal{D})$ has property $\mathcal{D}(s, c_1)$ with $s = p'$ and $c_1 = 1/p$; see Proposition 4.12 below for a more general class of such examples.

REMARK 2.11. Observe that the property $\mathcal{D}(s, c_1)$ can also be written as

$$|F_f(\varphi)| \leq c_1^{-1} \left(1 - \text{dist}(f, [\varphi])\right)^{1/s}, \quad \forall \|f\| = 1, \quad \varphi \in \mathcal{D}.$$

One may ask whether this property could hold for some non-uniformly smooth Banach space $(\mathbb{X}, \|\cdot\|)$ and some dictionary $\mathcal{D}$.

PROOF INTUITION

The obstruction is that an ℓ_1 direct sum has two independent norm components. Given any proposed unit atom $\varphi = (u, v)$, one can build vectors x_t whose two components point in exactly the same directions as u and v , but with very different relative sizes. Every norming functional for x_t still evaluates φ with modulus one. However, because one component of x_t is much larger than the matching component of the atom, subtracting a scalar multiple of φ cannot reduce the norm by a fixed proportion. The best distance ratio tends to one, contradicting any inequality with a fixed positive c_1 .

THEOREM

Let E and F be nonzero Banach spaces over $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, and let

$$X = E \oplus_1 F, \quad \|(e, f)\| = \|e\| + \|f\|.$$

Then for every unit vector $\varphi \in X$, every $s > 1$, and every $c > 0$, the inequality

$$\text{dist}(x, [\varphi]) \leq \|x\|(1 - c|G_x(\varphi)|^s)$$

fails for some $x \neq 0$, no matter how one chooses associated norming functionals G_x . Consequently no nonempty dictionary in a nontrivial ℓ_1 sum can satisfy property $\mathcal{D}(s, c)$.

PROOF

Let $\varphi = (u, v) \in X$ with $\|\varphi\| = 1$. Put

$$A = \|u\|, \quad B = \|v\|,$$

so $A + B = 1$.

First suppose $A, B > 0$. Set $e = u/A$ and $y = v/B$, so $\|e\| = \|y\| = 1$. If $A \leq B$, take

$$x_t = (e, ty), \quad t > 0.$$

Then $\|x_t\| = 1 + t$. Any associated norming functional for x_t has the form $G_t = (e_t^*, y_t^*)$, with $\|e_t^*\| = \|y_t^*\| = 1$, $e_t^*(e) = 1$, and $y_t^*(y) = 1$. Hence

$$G_t(\varphi) = e_t^*(u) + y_t^*(v) = A + B = 1.$$

For any scalar λ ,

$$\|x_t - \lambda\varphi\| = |1 - \lambda A| + |t - \lambda B|.$$

Since $A/B \leq 1$,

$$|1 - \lambda A| + |t - \lambda B| \geq |1 - \lambda A| + \frac{A}{B}|t - \lambda B| \geq 1 - \frac{A}{B}t.$$

Equality is attained by taking $\lambda = t/B$. Therefore

$$\text{dist}(x_t, [\varphi]) = 1 - \frac{A}{B}t.$$

Dividing by $\|x_t\| = 1 + t$ and letting $t \downarrow 0$ gives

$$\frac{\text{dist}(x_t, [\varphi])}{\|x_t\|} \longrightarrow 1, \quad |G_t(\varphi)| = 1.$$

Thus no fixed $c > 0$ can make the desired inequality true for all t .

If $A > B$, the same argument with $x_t = (te, y)$ gives

$$\text{dist}(x_t, [\varphi]) = 1 - \frac{B}{A}t, \quad \|x_t\| = 1 + t, \quad |G_t(\varphi)| = 1,$$

again contradicting any fixed positive c .

It remains to consider the pure cases. If $A = 0$, then $B = 1$. Choose any unit vector $e \in E$, put $y = v$, and set $x_t = (e, ty)$. Every norming functional for x_t evaluates φ as 1, while

$$\text{dist}(x_t, [\varphi]) = 1, \quad \|x_t\| = 1 + t.$$

Letting $t \downarrow 0$ again contradicts any positive c . The case $B = 0$ is symmetric. This proves the atom-level obstruction for every unit φ , and therefore for every nonempty dictionary. $\square$

LIMITATIONS AND NOVELTY CHECK

The theorem is only a partial negative result. It does not prove that property $\mathcal{D}(s, c_1)$ forces uniform smoothness, and it does not rule out all non-uniformly smooth norms or all possible thin dictionaries. It shows that non-uniform smoothness coming from a nontrivial ℓ_1 direct-sum corner is incompatible with the property at the level of every single atom.

The run indexes were searched for arXiv:1909.13536, the paper title, $\mathcal{D}(s, c_1)$, non-uniformly smooth Banach spaces, and WCGA Lebesgue-type terms. A web search found the source arXiv page but no later exact answer to Remark 2.11; a Semantic Scholar API citation query was rate-limited. Novelty confidence is therefore modest and should be checked by a human reviewer.

REFERENCES

- [1] S. Dilworth, G. Garrigós, E. Hernández, D. Kutzarova, and V. Temlyakov, *Lebesgue-type inequalities in greedy approximation*, arXiv:1909.13536.